

|               |                                                                     |
|---------------|---------------------------------------------------------------------|
| Assignment No | 14                                                                  |
| Title         | A) Teacher Application<br>B) Custom Tag Application                 |
| Objective     | A) Adding, Deleting and Viewing Records.<br>B) Creating Custom Tags |
| Roll No       | MCA2511                                                             |

A) Teacher Application

**Code:**

**index.jsp**

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"%>
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<meta charset="ISO-8859-1">
```

```
<title>Teacher Details</title>
```

```
<style>
```

```
    body {
        font-family: Arial, Helvetica, sans-serif;
        background-color: #f4f6f9;
        margin: 0;
        padding: 0;
    }
```

```
    .container {
        width: 400px;
        margin: 80px auto;
        background: #ffffff;
        padding: 25px 30px;
        border-radius: 8px;
        box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
    }
```

```
.container h2 {
    text-align: center;
    margin-bottom: 20px;
    color: #333;
}

.form-group {
    margin-bottom: 15px;
}

.form-group label {
    display: block;
    margin-bottom: 5px;
    font-weight: bold;
    color: #555;
}

.form-group input {
    width: 100%;
    padding: 10px;
    font-size: 14px;
    border: 1px solid #ccc;
    border-radius: 4px;
    box-sizing: border-box;
}

.form-group input:focus {
    border-color: #007bff;
    outline: none;
}

.btn-submit {
    width: 100%;
    padding: 10px;
    background-color: #007bff;
    border: none;
    border-radius: 4px;
    color: #ffffff;
    font-size: 16px;
    cursor: pointer;
    transition: background-color 0.3s ease;
}

.btn-submit:hover {
    background-color: #0056b3;
}
</style>
</head>
```

```
<body>

<div class="container">
  <h2>Add New Teacher</h2>

  <form action="AddRecord.jsp" method="post">

    <div class="form-group">
      <label for="tid">Teacher ID</label>
      <input type="number" id="tid" name="tid" required>
    </div>

    <div class="form-group">
      <label for="tname">Teacher Name</label>
      <input type="text" id="tname" name="tname" required>
    </div>

    <div class="form-group">
      <label for="expr">Experience (Years)</label>
      <input type="number" id="expr" name="expr" required>
    </div>

    <div class="form-group">
      <label for="specialization">Specialization</label>
      <input type="text" id="specialization" name="specialization" required>
    </div>

    <input type="submit" class="btn-submit" value="Add Teacher">

  </form>
</div>
<div class="container">
  <h2>Delete Teacher</h2>

  <form action="Delete.jsp" method="post">

    <div class="form-group">
      <label for="tid">Teacher ID</label>
      <input type="number" id="tid" name="tid" required>
    </div>

    <input type="submit" class="btn-submit" value="Delete Teacher">

  </form>
</div>

<div class="container">
```

<h2>View Teacher</h2>

<form action="TeacherDetails.jsp" method="post">

<input type="submit" class="btn-submit" value="View Teacher">

</form>

</div>

</body>

</html>

### AddRecord.jsp

<%@ page language="java" contentType="text/html; charset=ISO-8859-1"  
pageEncoding="ISO-8859-1"%>

<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<%@ taglib uri="http://java.sun.com/jsp/jstl/sql" prefix="sql" %>

<%@ taglib uri="http://java.sun.com/jsp/jstl/fmt" prefix="fmt" %>

<!DOCTYPE html>

<html>

<body>

<!-- 1. Configure the datasource -->

<sql:setDataSource

var="myds" driver="org.postgresql.Driver"

url="jdbc:postgresql://localhost:5432/postgres"

user="postgres" password="password" scope="application"/>

<!-- 2. Fetch the data from index.jsp using jstl -->

<fmt:parseNumber var="id" value="{param.tid}"/>

<fmt:parseNumber var="exp" value="{param.expr}"/>

<!-- 3. Configure the Query -->

<sql:update dataSource="{myds}" var="updq">

INSERT INTO TEACHERS VALUES (?, ?, ?, ?);

<sql:param value="{id}"/>

<sql:param value="{param.tname}"/>

<sql:param value="{exp}"/>

<sql:param value="{param.specialization}"/>

</sql:update>

<c:if test="{updq}>0">

<c:out value="Inserted Successfully"/>

</c:if>

</body>

</html>

### Delete.jsp

<!-- Pending -->

<%@ page language="java" contentType="text/html; charset=ISO-8859-1"  
pageEncoding="ISO-8859-1"%>

<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<%@ taglib uri="http://java.sun.com/jsp/jstl/sql" prefix="sql" %>

<%@ taglib uri="http://java.sun.com/jsp/jstl/fmt" prefix="fmt" %>

<!DOCTYPE html>

<html>

<body>

<!-- 1. Configure the datasource -->

<sql:setDataSource

var="myds" driver="org.postgresql.Driver"

url="jdbc:postgresql://localhost:5432/postgres"

user="postgres" password="password" scope="application"/>

<!-- 2. Fetch the data from index.jsp using jstl -->

<fmt:parseNumber var="id" value="{param.tid}"/>

<!-- 3. Configure the Query -->

<sql:update dataSource="{myds}" var="delq">

DELETE FROM TEACHERS WHERE tid=?

<sql:param value="{id}" />

</sql:update>

<c:if test="{delq}>0">

<c:out value="Deleted Successfully"/>

</c:if>

</body>

</html>

### TeacherDetails.jsp

<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<%@ taglib uri="http://java.sun.com/jsp/jstl/sql" prefix="sql" %>

<%@ page language="java" %>

<!DOCTYPE html>

<html>

<head>

```
<meta charset="ISO-8859-1">
<title>Teacher Details</title>
</head>
<body>
    <sql:setDataSource var="myds" driver="org.postgresql.Driver"
url="jdbc:postgresql://localhost:5432/postgres" user="postgres" password="password"
scope="application"/>
    <sql:query var="myque" dataSource="{myds}">select * from teachers</sql:query>
    <table border=1>
        <tr>
            <th>Teacher ID</th>
            <th>Teacher Name</th>
            <th>Experience</th>
            <th>Specialization</th>
        </tr>
        <c:forEach var="result" items="{myque.rows}">
            <tr>
                <td><c:out value="{result.tid}"/></td>
                <td><c:out value="{result.tname}"/></td>
                <td><c:out value="{result.expr}"/></td>
                <td><c:out value="{result.specialization}"/></td>
            </tr>
        </c:forEach>
    </table>
</body>
</html>
```

Output:

## Add New Teacher

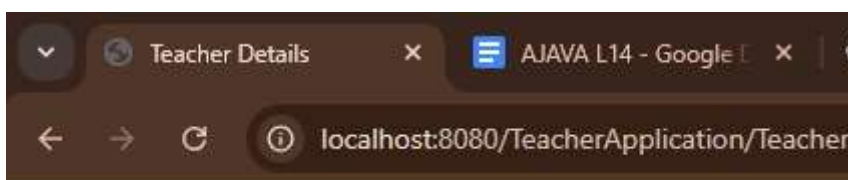
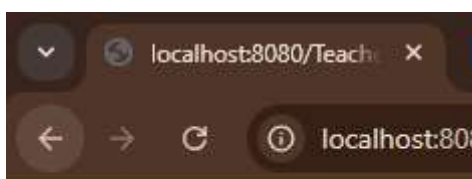
**Teacher ID**

**Teacher Name**

**Experience (Years)**

**Specialization**

**Add Teacher**

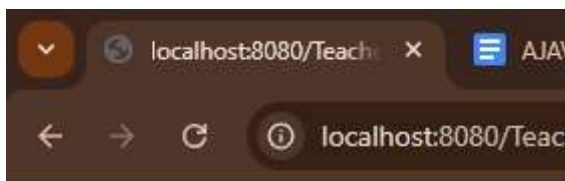


| Teacher ID | Teacher Name | Experience | Specialization |
|------------|--------------|------------|----------------|
| 1          | Mukund       | 5 years    | Spring         |
| 2          | Jay          | 8          | C++            |

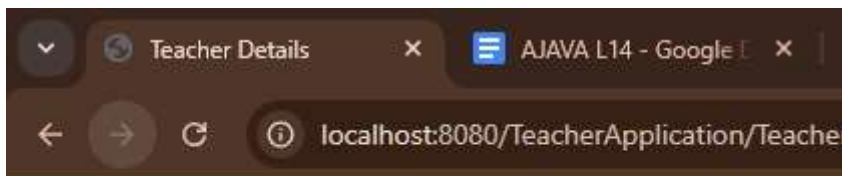
## Delete Teacher

Teacher ID

Delete Teacher



Deleted Successfully



| Teacher ID | Teacher Name | Experience | Specialization |
|------------|--------------|------------|----------------|
| 1          | Mukund       | 5 years    | Spring         |

## View Teacher

View Teacher



## B) Creating Custom Tags

### Code:

#### CustomDemo.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"%>

<!-- Import the Tag lib -->
<%@ taglib uri="/WEB-INF/tlds/myTags.tdl" prefix="mytag" %>

<!DOCTYPE html>
<html>
<body>
    <mytag:met/>
    <mytag:datetag/>
</body>
</html>
```

#### MyTagHandler.java

```
package edu.met.p1;

import java.io.IOException;

import javax.servlet.jsp.JspException;
import javax.servlet.jsp.JspWriter;
import javax.servlet.jsp.tagext.SimpleTagSupport;

public class MyTagHandler extends SimpleTagSupport {

    @Override
    public void doTag() throws JspException, IOException
    {
        //purpose is to print message
        JspWriter out = getJspContext().getOut();
        out.println("<h1>Mumbai Educational Trust</h1>");
    }
}
```

#### DateTag.java

```
package edu.met.p1;

import java.io.IOException;
```

```
import javax.servlet.jsp.JspException;
import javax.servlet.jsp.JspWriter;
import javax.servlet.jsp.tagext.TagSupport;

public class DateTag extends TagSupport
{
    @Override
    public int doStartTag() throws JspException
    {
        try
        {
            pageContext.getOut().println("Current Date and Time");
        } catch (Exception ex) { }
        return SKIP_BODY;
    }

    public int doEndTag() throws JspException
    {
        try
        {
            pageContext.getOut().println(new java.util.Date());
        } catch (Exception ex) { }
        return EVAL_PAGE;
    }
}
```

### **myTags.tdl**

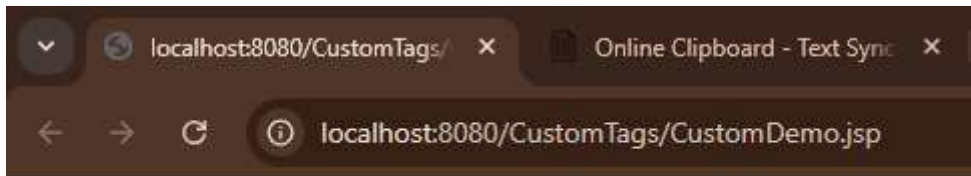
```
<?xml version="1.0" encoding="UTF-8"?>
<taglib xmlns="http://www.oracle.com/technetwork/java/index.html"
        xmlns:xsi="http://www.w3.org"

xsi:schemaLocation="http://www.oracle.com/webfolder/technetwork/jsc/xml/ns/j2ee/web-jsp
taglibrary_2_0.xsd"
        version="2.0">

    <tlib-version>1.0</tlib-version>
    <short-name>mytags</short-name>
    <uri>/WEB-INF/tlds/myTags.tdl</uri>
    <tag>
        <name>met</name>
        <tag-class>edu.met.p1.MyTagHandler</tag-class>
        <body-content>This is Empty</body-content>
    </tag>
    <tag>
        <name>datetag</name>
```

```
<tag-class>edu.met.pl.DateTag</tag-class>  
<body-content>Current Date</body-content>  
</tag>  
</taglib>
```

**Output:**



## Mumbai Educational Trust

Current Date and Time Thu Dec 18 11:06:08 IST 2025